

WasteWise System

Waste Collection Monitoring and Management Platform for Ormoc City

Overview

WasteWise is a waste collection monitoring and management platform developed for Ormoc City. The system provides real-time monitoring, route management, truck tracking, attendance monitoring, reporting, and complaint management for waste collection operations.

The platform is accessible through both web and mobile applications and supports the following users:

- System Administrators
- ENRO Staff
- Barangay Officials
- Garbage Collectors
- Residents

System Architecture

The WasteWise platform consists of the following components:

Web Application

- Hosted on Vercel
- Used by System Administrators, ENRO Staff, and Barangay Officials

Backend Server

- Hosted on Render
- Handles:
 - API requests
 - Database operations
 - Authentication and OTP verification
 - Real-time WebSocket communication
 - Data synchronization
 - Report and complaint management

Mobile Application

- Developed using React Native
- Used by Garbage Collectors and Residents

Real-Time Communication

- Powered by WebSocket technology
- Enables live updates for:
 - Truck tracking
 - Route completion

- Attendance monitoring
 - Notifications
 - Reports and complaints
-

System Requirements

Web Application Requirements

Supported Browsers

- Google Chrome (Latest Version)
- Microsoft Edge (Latest Version)

Minimum Requirements

- Desktop or laptop computer
 - Stable internet connection
 - Updated web browser
-

Mobile Application Requirements

Supported Platform

- Android

Minimum Requirements

- Android 8.0 (Oreo) or later
 - Minimum 3 GB RAM
 - At least 200 MB free storage
 - Stable internet connection
-

Web Application Access

Official Website

<https://waste-wise-ormoc-ddm.vercel.app/>

Login Procedure

1. Open your preferred web browser.
2. Navigate to the WasteWise website.
3. Enter your registered email address and password.
4. Click **Login**.
5. Check your email for the One-Time Password (OTP).
6. Enter the OTP.

7. Click **Verify**.
 8. The system dashboard will be displayed.
-

Account Access Request

Available for:

- ENRO Staff
- Barangay Officials

Steps

1. Open the login page.
2. Click **Request Account Access**.
3. Fill in the required information:
4. Full Name
5. Gender
6. Mobile Number
7. Email Address
8. Role(s)
9. Click **Submit Request**.

Approval Process

- The request is reviewed by the System Administrator.
 - Once approved, a notification is sent via email.
 - Login using the approved credentials and complete OTP verification.
-

Password Recovery (Web Application)

Steps

1. Click **Forgot Password** on the login page.
 2. Enter your registered email address.
 3. Click **Send Code**.
 4. Check your email for the OTP.
 5. Enter the OTP.
 6. Create a new password.
 7. Click **Save Password**.
 8. Login using your new credentials.
-

Mobile Application User Guide

The mobile application supports two user roles:

- Garbage Collector
 - Resident
-

Garbage Collector User Manual

Login

Buttons Used

- Login
- Verify OTP

Steps

1. Open the WasteWise mobile application.
 2. Enter your email and password.
 3. Tap **Login**.
 4. Check your email for the OTP.
 5. Enter the OTP.
 6. Tap **Verify OTP**.
-

Dashboard

The Dashboard provides a summary of:

- Daily assigned route
 - Attendance status
 - Collection progress
 - Notifications
-

Attendance

Buttons Used

- Clock In
- Clock Out

Clock In Procedure

1. Open the Attendance page.
2. Tap **Clock In**.

3. Confirm the action.

System Response

- Status changes to **On Duty**.
- Collection operations become visible to ENRO Staff and Residents.
- The Clock Out button becomes available.

Clock Out Procedure

1. Open the Attendance page.
2. Tap **Clock Out**.
3. Confirm the action.

System Response

- Attendance is recorded.
 - Status changes to **Off Duty**.
-

Schedule

Button Used

- View Schedule

Steps

1. Open the Schedule page.
 2. View assigned routes and collection schedules.
 3. Review upcoming collection assignments.
-

Route Management

Buttons Used

- Map View
- Check

Steps

1. Open the Route page.
2. Follow the route displayed on the map.
3. Proceed from the Start Point to the End Point.
4. After completing a collection point, tap **Check**.

System Response

- Collection point status changes from **Pending** to **Completed**.
 - Updates are reflected in real time.
-

Report Submission

Buttons Used

- Report
- Submit Report

Steps

1. Open the Report page.
2. Tap **Report**.
3. Select a report category:
 4. Vehicle Issue
 5. Equipment Problem
 6. Route Issue
 7. Safety Incident
8. Select the specific issue.
9. Enter remarks or additional information.
10. Tap **Submit Report**.

System Response

- ENRO Staff receive immediate notification.
-

Settings

Available Actions

- Edit Profile
- View Login History

Features

- Update personal information
 - View login records
 - Monitor account activity
-

Resident User Manual

Track Page

Buttons Used

- Search
- Filter

Steps

1. Open the WasteWise mobile application.
 2. The Track page loads automatically.
 3. Use Search to find:
 4. Truck ID
 5. Route Name
 6. Barangay
 7. Use Filters to narrow results:
 8. Truck Status
 9. Garbage Type
 10. Collection Schedule
 11. View real-time truck locations on the map.
-

Collection Logs

Buttons Used

- View Logs
- Filter

Steps

1. Open the Logs page.
 2. Review collection records.
 3. View status:
 4. Completed
 5. Incomplete
 6. Apply filters as needed.
-

Complaint Reporting

Button Used

- Submit Complaint

Steps

1. Open the Report page.
2. Tap **Submit Complaint**.

3. Select a complaint type:
4. Uncollected Garbage
5. Overflowing Garbage
6. Illegal Dumping
7. Missed Route
8. Others
9. Select the garbage type.
10. Enter additional notes.
11. Tap **Submit**.

System Response

- Complaint is sent directly to ENRO Staff for review.
-

Password Recovery (Mobile App)

Steps

1. Tap **Forgot Password** on the login screen.
 2. Enter your registered email address.
 3. Tap **Send Code**.
 4. Enter the OTP received via email.
 5. Create a new password.
 6. Tap **Save**.
-

System Administrator User Manual

User Management

Buttons Used

- Add New User
- Save

Steps

1. Open User Management.
2. Click **Add New User**.
3. Enter user details.
4. Assign appropriate roles.
5. Click **Save**.

System Response

- User credentials are generated and sent via email.
-

Role Action Management

Buttons Used

- Add Role Action
- Save

Steps

1. Open Role Action Management.
 2. Click **Add Role Action**.
 3. Select a role.
 4. Configure permissions and capabilities.
 5. Click **Save**.
-

ENRO Staff User Manual

Barangay Management

Buttons Used

- Barangay Management
- Add Barangay
- Save

Functions

- View all barangays
 - Add new barangays
 - Update barangay records
-

Truck Management

Buttons Used

- Truck Management
- Add New Truck
- Save

Information Required

- Truck ID
- Truck Status
- Assigned Garbage Collector

Available Statuses

- Active

- On Route
 - Inactive
 - Under Maintenance
-

Route Management

Buttons Used

- Add New Route
- Add
- Save Route
- Clear Route

Steps

1. Create a route name.
2. Select barangay coverage.
3. Choose route color.
4. Define start point.
5. Draw route path on the map.
6. Define end point.
7. Save the route.

Purpose

Routes define official collection paths followed by garbage trucks.

Schedule Management

Buttons Used

- Add New Schedule
- Save

Steps

1. Select a route.
 2. Assign a garbage collector.
 3. Verify truck assignment.
 4. Select garbage type.
 5. Choose collection days.
 6. Save the schedule.
-

Garbage Report Management

Buttons Used

- View
- Update Status

Functions

- Review resident complaints
 - Track complaint progress
 - Update status:
 - Pending
 - In Progress
 - Resolved
-

Garbage Site Management

Buttons Used

- Add New Garbage Site
- Save

Steps

1. Select a barangay.
2. Enter site name.
3. Mark the drop-off location on the map.
4. Save the site.

Purpose

Provides residents with designated garbage drop-off locations.

Collector Report Management

Buttons Used

- View
- Update Status

Functions

- Review collector-submitted reports
 - Investigate issues
 - Update resolution status
-

Collector Attendance Management

Button Used

- View

Functions

Monitor collector attendance:

- Clocked In
- Clocked Out

Used for workforce and operational monitoring.

Truck Map Monitoring

Buttons Used

- Search
- Filter

Functions

View all active trucks and routes in real time.

Search Criteria

- Truck ID
- Route
- Barangay

Filter Options

- Truck Status
 - Route
 - Barangay
-

Profile and Login History

Available Actions

- Edit Profile
- Change Password
- Login History

Features

- Update personal information

- Change account password
 - View login timestamps
 - View device information
 - Monitor login activity
-

Backend Deployment Information

The WasteWise Backend Server is deployed on Render.

Responsibilities

- User Authentication
- OTP Verification
- API Processing
- Database Management
- Real-Time Communication
- Attendance Monitoring
- Route Tracking
- Complaint Management
- Notification Delivery

The backend acts as the central communication layer between the web application, mobile application, and database services.

Real-Time Features

WasteWise supports real-time updates using WebSocket technology.

Supported Real-Time Functions

- Live Truck Tracking
 - Route Progress Monitoring
 - Attendance Updates
 - Complaint Notifications
 - Report Notifications
 - Schedule Synchronization
-

© 2025 WasteWise – Waste Collection Monitoring and Management System for Ormoc City.